

Package ‘manCTMed’

July 30, 2026

Title Continuous Time Mediation

Version 1.0.9

Description Research compendium for the manuscript
Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025).
Inferences and Effect Sizes for Direct, Indirect, and Total Effects
in Continuous-Time Mediation Models.
Psychological Methods.
<[doi:10.1037/met0000779](https://doi.org/10.1037/met0000779)>.

URL <https://github.com/jeksterslab/manCTMed>,
<https://jeksterslab.github.io/manCTMed/>, <https://osf.io/qwnmf/>,
<https://doi.org/10.1037/met0000779>

BugReports <https://github.com/jeksterslab/manCTMed/issues>

License MIT + file LICENSE

Encoding UTF-8

LazyData true

LazyDataCompression xz

Roxygen list(markdown = TRUE)

Depends R (>= 4.0.0)

Imports stats, dynr, OpenMx, dynUtils, simStateSpace (== 1.2.9),
bootStateSpace (== 1.0.2), cTMed (== 1.0.6), ggplot2

Suggests knitr, rmarkdown, testthat, DT

Remotes jeksterslab/dynUtils@dc3f47b

Config/roxygen2/version 8.0.0.9000

NeedsCompilation no

Author Ivan Jacob Agaloos Pesigan [aut, cre, cph] (ORCID:
<<https://orcid.org/0000-0003-4818-8420>>),
Michael A. Russell [ctb] (ORCID:
<<https://orcid.org/0000-0002-3956-604X>>),
Sy-Miin Chow [ctb] (ORCID: <<https://orcid.org/0000-0003-1938-027X>>)

Maintainer Ivan Jacob Agaloos Pesigan <r.jeksterslab@gmail.com>

Contents

BootPara	4
BootParaStdXMY	5
BootParaStdXYM	6
BootParaStdYMX	7
BootParaXMY	8
BootParaXYM	9
BootParaYMX	10
Compress	11
DeltaStdXMY	12
DeltaStdXYM	12
DeltaStdYMX	13
DeltaXMY	14
DeltaXYM	15
DeltaYMX	16
FigPlotEffects	16
FigScatterPlotCoverage	17
FigScatterPlotPower	18
FigScatterPlotSeBias	19
FigScatterPlotType1	20
FitDynr	21
FitMx	21
GenData	22
IllustrationBootPara	23
IllustrationCompress	24
IllustrationFigPlotEffects	25
IllustrationFigScatterPlotCoverage	25
IllustrationFigScatterPlotPower	26
IllustrationFigScatterPlotSeBias	27
IllustrationFitDynr	28
IllustrationFitMx	28
IllustrationGenData	29
IllustrationMCPhiSigma	30
IllustrationPrepData	30
illustration_dist	31
illustration_dist_dt	32
illustration_dist_dt_mc	32
illustration_dist_mc	33
illustration_dist_med	33
illustration_dist_med_mc	34
illustration_dist_med_std	34
illustration_dist_med_std_mc	35
illustration_results	35
MCStdXMY	37
MCStdXYM	38
MCStdYMX	39
MCXMY	40

MCXYM	41
MCYMX	42
params	43
PhiHat	43
RandomMeasurement	44
results	45
Sim	46
SimDynrBootPara	47
SimDynrBootParaStdXMY	48
SimDynrBootParaStdYMX	49
SimDynrBootParaXMY	50
SimDynrBootParaYMX	51
SimDynrDeltaStdXMY	52
SimDynrDeltaStdYMX	53
SimDynrDeltaXMY	54
SimDynrDeltaYMX	55
SimDynrMCStdXMY	56
SimDynrMCStdYMX	57
SimDynrMCXMY	58
SimDynrMCYMX	59
SimFitDynr	60
SimFitMx	61
SimFN	62
SimGenData	62
SimIllustration	63
SimIllustrationDynrBootPara	64
SimIllustrationDynrBootParaStdXMY	65
SimIllustrationDynrBootParaXMY	66
SimIllustrationDynrDeltaStdXMY	67
SimIllustrationDynrDeltaXMY	68
SimIllustrationDynrMCPhiSigma	69
SimIllustrationDynrMCStdXMY	70
SimIllustrationDynrMCXMY	71
SimIllustrationFitDynr	72
SimIllustrationFitMx	73
SimIllustrationGenData	74
SimIllustrationPara	75
SimPara	76
SimProj	76
Sum	77
SumDynrDeltaStdXMY	77
SumDynrDeltaStdYMX	78
SumDynrDeltaXMY	79
SumDynrDeltaYMX	80
SumDynrMCStdXMY	80
SumDynrMCStdYMX	81
SumDynrMCXMY	82
SumDynrMCYMX	83

SumFitDynr	83
SumIllustration	84
SumIllustrationDynrBootParaStdXMY	85
SumIllustrationDynrBootParaXMY	86
SumIllustrationDynrDeltaStdXMY	87
SumIllustrationDynrDeltaXMY	88
SumIllustrationDynrMCStdXMY	88
SumIllustrationDynrMCXMY	89
SumIllustrationFitDynr	90
ThetaHat	91

Index	92
--------------	-----------

BootPara	<i>Parametric Bootstrap</i>
----------	-----------------------------

Description

The function generates simulated datasets based on a fitted model and refits the model to each generated dataset using the dynr package.

Usage

```
BootPara(
  fit,
  path,
  prefix,
  taskid,
  B = 1000L,
  ncores = NULL,
  seed = NULL,
  clean = TRUE
)
```

Arguments

fit	R object. Output of the <code>FitDynr()</code> , <code>FitMx()</code> , <code>IllustrationFitDynr()</code> , or <code>IllustrationFitMx()</code> , functions.
path	Path to a directory to store bootstrap samples and estimates.
prefix	Character string. Prefix used for the file names for the bootstrap samples and estimates.
taskid	Positive integer. Task ID.
B	Positive integer. Number of bootstrap samples.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.
clean	Logical. If <code>clean = TRUE</code> , delete intermediate files generated by the function.

See Also

Other Confidence Interval Functions: [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)

## End(Not run)
```

 BootParaStdXMY

Parametric Bootstrap Confidence Intervals for X-M-Y (Standardized)

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \rightarrow M \rightarrow Y$ (Standardized).

Usage

```
BootParaStdXMY(boot, theta_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)
theta_hat <- ThetaHat(fit)
ci <- BootParaStdXMY(boot = boot, theta_hat = theta_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

BootParaStdXYM

Parametric Bootstrap Confidence Intervals for X-Y-M (Standardized)

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \rightarrow Y \rightarrow M$ (Standardized).

Usage

```
BootParaStdXYM(boot, theta_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)
theta_hat <- ThetaHat(fit)
ci <- BootParaStdXYM(boot = boot, theta_hat = theta_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

BootParaStdYMX

Parametric Bootstrap Confidence Intervals for Y-M-X (Standardized)

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $Y \rightarrow M \rightarrow X$ (Standardized).

Usage

```
BootParaStdYMX(boot, theta_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)
theta_hat <- ThetaHat(fit)
ci <- BootParaStdYMX(boot = boot, theta_hat = theta_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

 BootParaXMY

Parametric Bootstrap Confidence Intervals for X-M-Y

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \rightarrow M \rightarrow Y$.

Usage

```
BootParaXMY(boot, phi_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)
phi_hat <- PhiHat(fit)
ci <- BootParaXYM(boot = boot, phi_hat = phi_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

 BootParaXYM

Parametric Bootstrap Confidence Intervals for X-Y-M

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \rightarrow Y \rightarrow M$.

Usage

```
BootParaXYM(boot, phi_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)
phi_hat <- PhiHat(fit)
ci <- BootParaXYM(boot = boot, phi_hat = phi_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

 BootParaYMX

Parametric Bootstrap Confidence Intervals for Y-M-X

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $Y \rightarrow M \rightarrow X$.

Usage

```
BootParaYMX(boot, phi_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)
phi_hat <- PhiHat(fit)
ci <- BootParaYMX(boot = boot, phi_hat = phi_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

Compress

*Compress Replication***Description**

Compress Replication

Usage

```
Compress(taskid, repid, output_folder)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Compression Functions: [IllustrationCompress\(\)](#)

DeltaStdXMY

*Delta Method Confidence Intervals for X-M-Y (Standardized)***Description**

The function generates delta method confidence intervals for the mediation model $X \rightarrow M \rightarrow Y$ (Standardized).

Usage

```
DeltaStdXMY(theta_hat, delta_t = 1:30)
```

Arguments

theta_hat R object. Output of the [ThetaHat\(\)](#) function.
 delta_t Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- DeltaStdXMY(theta_hat)
plot(ci)

## End(Not run)
```

DeltaStdXYM

*Delta Method Confidence Intervals for X-Y-M (Standardized)***Description**

The function generates delta method confidence intervals for the mediation model $X \rightarrow Y \rightarrow M$ (Standardized).

Usage

```
DeltaStdXYM(theta_hat, delta_t = 1:30)
```

Arguments

theta_hat R object. Output of the [ThetaHat\(\)](#) function.

delta_t Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- DeltaStdXYM(theta_hat)
plot(ci)

## End(Not run)
```

DeltaStdYMX

Delta Method Confidence Intervals for Y-M-X (Standardized)

Description

The function generates delta method confidence intervals for the mediation model $Y \rightarrow M \rightarrow X$ (Standardized).

Usage

```
DeltaStdYMX(theta_hat, delta_t = 1:30)
```

Arguments

theta_hat R object. Output of the [ThetaHat\(\)](#) function.

delta_t Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- DeltaStdYMX(theta_hat)
plot(ci)

## End(Not run)
```

DeltaXMY

*Delta Method Confidence Intervals for X-M-Y***Description**

The function generates delta method confidence intervals for the mediation model $X \rightarrow M \rightarrow Y$.

Usage

```
DeltaXMY(phi_hat, delta_t = 1:30)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- DeltaXYM(phi_hat)
plot(ci)

## End(Not run)
```

DeltaXYM

*Delta Method Confidence Intervals for X-Y-M***Description**

The function generates delta method confidence intervals for the mediation model $X \rightarrow Y \rightarrow M$.

Usage

```
DeltaXYM(phi_hat, delta_t = 1:30)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- DeltaXYM(phi_hat)
plot(ci)

## End(Not run)
```

DeltaYMX

*Delta Method Confidence Intervals for Y-M-X***Description**

The function generates delta method confidence intervals for the mediation model $Y \rightarrow M \rightarrow X$.

Usage

```
DeltaYMX(phi_hat, delta_t = 1:30)
```

Arguments

phi_hat R object. Output of the [PhiHat\(\)](#) function.
 delta_t Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- DeltaYMX(phi_hat)
plot(ci)

## End(Not run)
```

FigPlotEffects

*Plot Total, Direct, and Indirect Effects***Description**

Effects for the model $X \rightarrow M \rightarrow Y$.

Usage

```
FigPlotEffects(dynamics = 0, std = FALSE, max_delta_t = 30, xmy = TRUE)
```


Arguments

dynamics	Integer. dynamics = 0 for original drift matrix, dynamics = -1 for near-neutral dynamics, and dynamics = 1 for stronger damping.
std	Logical. If std = TRUE, standardized total, direct, and indirect effects. If std = FALSE, unstandardized total, direct, and indirect effects.
max_delta_t	Numeric. Maximum time interval.
xmy	Logical. If xmy = TRUE, plot the effects for the $x \rightarrow m \rightarrow y$ mediation model. If xmy = FALSE, plot the effects for the $y \rightarrow m \rightarrow x$ mediation model.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigScatterPlotCoverage\(\)](#), [FigScatterPlotPower\(\)](#), [FigScatterPlotSeBias\(\)](#), [FigScatterPlotType1\(\)](#), [IllustrationFigPlotEffects\(\)](#), [IllustrationFigScatterPlotCoverage\(\)](#), [IllustrationFigScatterPlotPower\(\)](#), [IllustrationFigScatterPlotSeBias\(\)](#)

Examples

```
FigPlotEffects()
```

FigScatterPlotCoverage

Plot Coverage Probabilities

Description

Coverage probabilities for the model $X \rightarrow M \rightarrow Y$.

Usage

```
FigScatterPlotCoverage(results, delta_t = NULL, dynamics = 0, std = FALSE)
```

Arguments

results	Summary results data frame.
delta_t	Vector of time-interval value. If delta_t = NULL, use all available time-intervals
dynamics	Integer. dynamics = 0 for original drift matrix, dynamics = -1 for near-neutral dynamics, and dynamics = 1 for stronger damping.
std	Logical. If std = TRUE, standardized total, direct, and indirect effects. If std = FALSE, unstandardized total, direct, and indirect effects.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigPlotEffects\(\)](#), [FigScatterPlotPower\(\)](#), [FigScatterPlotSeBias\(\)](#), [FigScatterPlotType1\(\)](#), [IllustrationFigPlotEffects\(\)](#), [IllustrationFigScatterPlotCoverage\(\)](#), [IllustrationFigScatterPlotPower\(\)](#), [IllustrationFigScatterPlotSeBias\(\)](#)

Examples

```
data(results, package = "manCTMed")
FigScatterPlotCoverage(results)
FigScatterPlotCoverage(results, delta_t = 1:14)
FigScatterPlotCoverage(results, delta_t = 15:30)
```

FigScatterPlotPower *Plot Statistical Power*

Description

Statistical power for the model $X \rightarrow M \rightarrow Y$.

Usage

```
FigScatterPlotPower(results, delta_t = NULL, dynamics = 0, std = FALSE)
```

Arguments

results	Summary results data frame.
delta_t	Vector of time-interval value. If delta_t = NULL, use all available time-intervals
dynamics	Integer. dynamics = 0 for original drift matrix, dynamics = -1 for near-neutral dynamics, and dynamics = 1 for stronger damping.
std	Logical. If std = TRUE, standardized total, direct, and indirect effects. If std = FALSE, unstandardized total, direct, and indirect effects.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigPlotEffects\(\)](#), [FigScatterPlotCoverage\(\)](#), [FigScatterPlotSeBias\(\)](#), [FigScatterPlotType1\(\)](#), [IllustrationFigPlotEffects\(\)](#), [IllustrationFigScatterPlotCoverage\(\)](#), [IllustrationFigScatterPlotPower\(\)](#), [IllustrationFigScatterPlotSeBias\(\)](#)

Examples

```
data(results, package = "manCTMed")
FigScatterPlotPower(results)
FigScatterPlotPower(results, delta_t = 1:14)
FigScatterPlotPower(results, delta_t = 15:30)
```

FigScatterPlotSeBias *Plot Standard Error Bias*

Description

Standard Error Bias for the model $X \rightarrow M \rightarrow Y$.

Usage

```
FigScatterPlotSeBias(results, delta_t = NULL, dynamics = 0, std = FALSE)
```

Arguments

results	Summary results data frame.
delta_t	Vector of time-interval value. If delta_t = NULL, use all available time-intervals
dynamics	Integer. dynamics = 0 for original drift matrix, dynamics = -1 for near-neutral dynamics, and dynamics = 1 for stronger damping.
std	Logical. If std = TRUE, standardized total, direct, and indirect effects. If std = FALSE, unstandardized total, direct, and indirect effects.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigPlotEffects\(\)](#), [FigScatterPlotCoverage\(\)](#), [FigScatterPlotPower\(\)](#), [FigScatterPlotType1\(\)](#), [IllustrationFigPlotEffects\(\)](#), [IllustrationFigScatterPlotCoverage\(\)](#), [IllustrationFigScatterPlotPower\(\)](#), [IllustrationFigScatterPlotSeBias\(\)](#)

Examples

```
data(results, package = "manCTMed")
FigScatterPlotSeBias(results)
```

FigScatterPlotType1 *Plot Type I Error*

Description

Type I error for the model $Y \rightarrow M \rightarrow X$.

Usage

```
FigScatterPlotType1(results, delta_t = NULL, dynamics = 0, std = FALSE)
```

Arguments

results	Summary results data frame.
delta_t	Vector of time-interval value. If delta_t = NULL, use all available time-intervals
dynamics	Integer. dynamics = 0 for original drift matrix, dynamics = -1 for near-neutral dynamics, and dynamics = 1 for stronger damping.
std	Logical. If std = TRUE, standardized total, direct, and indirect effects. If std = FALSE, unstandardized total, direct, and indirect effects.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigPlotEffects\(\)](#), [FigScatterPlotCoverage\(\)](#), [FigScatterPlotPower\(\)](#), [FigScatterPlotSeBias\(\)](#), [IllustrationFigPlotEffects\(\)](#), [IllustrationFigScatterPlotCoverage\(\)](#), [IllustrationFigScatterPlotPower\(\)](#), [IllustrationFigScatterPlotSeBias\(\)](#)

Examples

```
data(results, package = "manCTMed")
FigScatterPlotType1(results)
FigScatterPlotType1(results, delta_t = 1:14)
FigScatterPlotType1(results, delta_t = 15:30)
```

Description

The function fits the model using the [dynr](#) package.

Usage

```
FitDynr(data, taskid)
```

Arguments

data	R object. Output of the RandomMeasurement() function.
taskid	Positive integer. Task ID.

See Also

Other Model Fitting Functions: [FitMx\(\)](#), [IllustrationFitDynr\(\)](#), [IllustrationFitMx\(\)](#), [IllustrationMCPhiSigma\(\)](#), [IllustrationPrepData\(\)](#), [PhiHat\(\)](#), [ThetaHat\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
summary(fit)

## End(Not run)
```

Description

The function fits the model using the [OpenMx](#) package.

Usage

```
FitMx(data, taskid)
```

Arguments

`data` R object. Output of the [RandomMeasurement\(\)](#) function.
`taskid` Positive integer. Task ID.

See Also

Other Model Fitting Functions: [FitDynr\(\)](#), [IllustrationFitDynr\(\)](#), [IllustrationFitMx\(\)](#), [IllustrationMCPHiSigma\(\)](#), [IllustrationPrepData\(\)](#), [PhiHat\(\)](#), [ThetaHat\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(OpenMx)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitMx(data, taskid = 1)
summary(fit)

## End(Not run)
```

GenData

Simulate Data

Description

The function simulates data using the [simStateSpace::SimSSMOUFixed\(\)](#) function.

Usage

```
GenData(taskid)
```

Arguments

`taskid` Positive integer. Task ID.

See Also

Other Data Generation Functions: [IllustrationGenData\(\)](#), [RandomMeasurement\(\)](#)

Examples

```
## Not run:
set.seed(42)
sim <- GenData(taskid = 1)
plot(sim)

## End(Not run)
```

IllustrationBootPara *Parametric Bootstrap (Illustration)*

Description

The function generates simulated datasets based on a fitted model and refits the model to each generated dataset using the dynr package.

Usage

```
IllustrationBootPara(
  fit,
  path,
  prefix,
  taskid,
  B = 1000L,
  ncores = NULL,
  seed = NULL
)
```

Arguments

fit	R object. Fitted CT-VAR model.
path	Path to a directory to store bootstrap samples and estimates.
prefix	Character string. Prefix used for the file names for the bootstrap samples and estimates.
taskid	Positive integer. Task ID.
B	Positive integer. Number of bootstrap samples.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
library(dynr)
sim <- IllustrationGenData(seed = 42)
data <- IllustrationPrepData(sim)
fit <- IllustrationFitDynr(data)
summary(fit)
```

```

IllustrationBootPara(
    fit = fit,
    path = getwd(),
    prefix = "pb",
    taskid = 1,
    B = 1000L,
    seed = 42
)

## End(Not run)

```

IllustrationCompress *Compress Replication (Illustration)*

Description

Compress Replication (Illustration)

Usage

```
IllustrationCompress(taskid, repid, output_folder)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Compression Functions: [Compress\(\)](#)

IllustrationFigPlotEffects

Plot Total, Direct, and Indirect Effects (Illustration)

Description

Effects for the model $X \rightarrow M \rightarrow Y$.

Usage

```
IllustrationFigPlotEffects(std = FALSE, max_delta_t = 30)
```

Arguments

std	Logical. If std = TRUE, standardized total, direct, and indirect effects. If std = FALSE, unstandardized total, direct, and indirect effects.
max_delta_t	Numeric. Maximum time interval.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigPlotEffects\(\)](#), [FigScatterPlotCoverage\(\)](#), [FigScatterPlotPower\(\)](#), [FigScatterPlotSeBias\(\)](#), [FigScatterPlotType1\(\)](#), [IllustrationFigScatterPlotCoverage\(\)](#), [IllustrationFigScatterPlotPower\(\)](#), [IllustrationFigScatterPlotSeBias\(\)](#)

Examples

```
IllustrationFigPlotEffects(std = FALSE)
IllustrationFigPlotEffects(std = TRUE)
```

IllustrationFigScatterPlotCoverage

Illustration Plot Coverage Probabilities

Description

Coverage probabilities for the model $X \rightarrow M \rightarrow Y$.

Usage

```
IllustrationFigScatterPlotCoverage(illustration_results)
```

Arguments

`illustration_results`
Summary results data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigPlotEffects\(\)](#), [FigScatterPlotCoverage\(\)](#), [FigScatterPlotPower\(\)](#), [FigScatterPlotSeBias\(\)](#), [FigScatterPlotType1\(\)](#), [IllustrationFigPlotEffects\(\)](#), [IllustrationFigScatterPlotCoverage\(\)](#), [IllustrationFigScatterPlotPower\(\)](#), [IllustrationFigScatterPlotSeBias\(\)](#)

Examples

```
data(illustration_results, package = "manCTMed")
IllustrationFigScatterPlotCoverage(illustration_results)
```

IllustrationFigScatterPlotPower
Illustration Plot Statistical Power

Description

Statistical Power for the model $X \rightarrow M \rightarrow Y$.

Usage

```
IllustrationFigScatterPlotPower(illustration_results)
```

Arguments

`illustration_results`
Summary results data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigPlotEffects\(\)](#), [FigScatterPlotCoverage\(\)](#), [FigScatterPlotPower\(\)](#), [FigScatterPlotSeBias\(\)](#), [FigScatterPlotType1\(\)](#), [IllustrationFigPlotEffects\(\)](#), [IllustrationFigScatterPlotCoverage\(\)](#), [IllustrationFigScatterPlotPower\(\)](#), [IllustrationFigScatterPlotSeBias\(\)](#)

Examples

```
data(illustration_results, package = "manCTMed")
IllustrationFigScatterPlotPower(illustration_results)
```

IllustrationFigScatterPlotSeBias

Illustration Plot Standard Error Bias

Description

Standard Error Bias for the model $X \rightarrow M \rightarrow Y$.

Usage

```
IllustrationFigScatterPlotSeBias(illustration_results)
```

Arguments

illustration_results
Summary results data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigPlotEffects\(\)](#), [FigScatterPlotCoverage\(\)](#), [FigScatterPlotPower\(\)](#),
[FigScatterPlotSeBias\(\)](#), [FigScatterPlotType1\(\)](#), [IllustrationFigPlotEffects\(\)](#), [IllustrationFigScatterPlotPower\(\)](#)

Examples

```
data(illustration_results, package = "manCTMed")
IllustrationFigScatterPlotSeBias(illustration_results)
```

IllustrationFitDynr	<i>Fit the Model using the dynr Package (Illustration)</i>
---------------------	--

Description

The function fits the model using the [dynr](#) package.

Usage

```
IllustrationFitDynr(data)
```

Arguments

data	R object. Output of the IllustrationPrepData() function.
------	--

See Also

Other Model Fitting Functions: [FitDynr\(\)](#), [FitMx\(\)](#), [IllustrationFitMx\(\)](#), [IllustrationMCPhiSigma\(\)](#), [IllustrationPrepData\(\)](#), [PhiHat\(\)](#), [ThetaHat\(\)](#)

Examples

```
## Not run:
library(dynr)
sim <- IllustrationGenData(seed = 42)
data <- IllustrationPrepData(sim)
fit <- IllustrationFitDynr(data)
summary(fit)

## End(Not run)
```

IllustrationFitMx	<i>Fit the Model using the OpenMx Package (Illustration)</i>
-------------------	--

Description

The function fits the model using the [OpenMx](#) package.

Usage

```
IllustrationFitMx(data)
```

Arguments

data	R object. Output of the IllustrationPrepData() function.
------	--

See Also

Other Model Fitting Functions: [FitDynr\(\)](#), [FitMx\(\)](#), [IllustrationFitDynr\(\)](#), [IllustrationMCPhiSigma\(\)](#), [IllustrationPrepData\(\)](#), [PhiHat\(\)](#), [ThetaHat\(\)](#)

Examples

```
## Not run:
library(OpenMx)
sim <- IllustrationGenData(seed = 42)
data <- IllustrationPrepData(sim)
fit <- IllustrationFitMx(data)
summary(fit)

## End(Not run)
```

IllustrationGenData	<i>Simulate Data (Illustration)</i>
---------------------	-------------------------------------

Description

The function simulates data using the [simStateSpace::SimSSMOUFixed\(\)](#) function.

Usage

```
IllustrationGenData(seed = NULL, n = 133, m = 101, delta_t_gen = 0.1)
```

Arguments

seed	Integer. Random seed.
n	Positive integer. Sample size.
m	Positive integer. Measurement occasions.
delta_t_gen	Numeric. Time interval used to generate data.

See Also

Other Data Generation Functions: [GenData\(\)](#), [RandomMeasurement\(\)](#)

Examples

```
## Not run:
sim <- IllustrationGenData(seed = 42)
plot(sim)

## End(Not run)
```

IllustrationMCPhiSigma

Generate a Sampling Distribution of Drift Matrices and Process Noise Covariance Matrices (Illustration)

Description

The function generates a sampling distribution of drift matrices and process noise covariance matrices using the Monte Carlo method.

Usage

```
IllustrationMCPhiSigma(fit, R = 20000L, seed = NULL)
```

Arguments

fit	R object. Fitted CT-VAR model.
R	Positive integer. Number of Monte Carlo replications.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDynr\(\)](#), [FitMx\(\)](#), [IllustrationFitDynr\(\)](#), [IllustrationFitMx\(\)](#), [IllustrationPrepData\(\)](#), [PhiHat\(\)](#), [ThetaHat\(\)](#)

Examples

```
## Not run:
library(dynr)
sim <- IllustrationGenData(seed = 42)
data <- IllustrationPrepData(sim)
fit <- IllustrationFitDynr(data)
IllustrationMCPhiSigma(fit, seed = 42)

## End(Not run)
```

IllustrationPrepData *Prepare Data Before Model Fitting (Illustration)*

Description

The function converts the output of [IllustrationGenData\(\)](#) into a data frame.

Usage

```
IllustrationPrepData(sim)
```

Arguments

`sim` R object. Output of the `IllustrationGenData()` function.

See Also

Other Model Fitting Functions: `FitDynr()`, `FitMx()`, `IllustrationFitDynr()`, `IllustrationFitMx()`, `IllustrationMCPhiSigma()`, `PhiHat()`, `ThetaHat()`

Examples

```
## Not run:
sim <- IllustrationGenData(seed = 42)
data <- IllustrationPrepData(sim)
head(data)
dim(data)

## End(Not run)
```

illustration_dist	<i>Illustration Sampling Distribution</i>
-------------------	---

Description

Illustration Sampling Distribution

Usage

```
data(illustration_dist)
```

Format

A matrix with 1000 rows and 27 columns:

phi_xx Elements of the drift matrix.

sigma_xx Elements of the process noise covariance matrix.

theta_xx Elements of the measurement error covariance matrix.

mu0_x Elements of the initial condition mean vector.

sigma0_xx Elements of the initial condition covariance matrix.

Author(s)

Ivan Jacob Agaloos Pesigan

illustration_dist_dt *Illustration Sampling Distribution Discrete Time - Time Interval of 1*

Description

Illustration Sampling Distribution Discrete Time - Time Interval of 1

Usage

```
data(illustration_dist_dt)
```

Format

A matrix with 1000 rows and 15 columns:

beta_xx Elements of the matrix of lagged coefficients.

psi_xx Elements of the process noise covariance matrix.

Author(s)

Ivan Jacob Agaloos Pesigan

illustration_dist_dt_mc
 Illustration Sampling Distribution Discrete Time - Time Interval of 1
 (Monte Carlo Method)

Description

Illustration Sampling Distribution Discrete Time - Time Interval of 1 (Monte Carlo Method)

Usage

```
data(illustration_dist_dt_mc)
```

Format

A matrix with 20000 rows and 15 columns:

beta_xx Elements of the matrix of lagged coefficients.

psi_xx Elements of the process noise covariance matrix.

Author(s)

Ivan Jacob Agaloos Pesigan

illustration_dist_mc *Illustration Sampling Distribution (Monte Carlo Method)*

Description

Illustration Sampling Distribution (Monte Carlo Method)

Usage

```
data(illustration_dist_mc)
```

Format

A matrix with 20000 rows and 15 columns:

phi_xx Elements of the drift matrix.

sigma_xx Elements of the process noise covariance matrix.

Author(s)

Ivan Jacob Agaloos Pesigan

illustration_dist_med *Illustration Sampling Distribution Total, Direct, and Indirect Effects - Time Interval of 1*

Description

Illustration Sampling Distribution Total, Direct, and Indirect Effects - Time Interval of 1

Usage

```
data(illustration_dist_med)
```

Format

A matrix with 1000 rows and 27 columns:

total Total effect.

direct Direct effect.

indirect Indirect effect.

Author(s)

Ivan Jacob Agaloos Pesigan

```
illustration_dist_med_mc
```

Illustration Sampling Distribution Total, Direct, and Indirect Effects - Time Interval of 1 (Monte Carlo Method)

Description

Illustration Sampling Distribution Total, Direct, and Indirect Effects - Time Interval of 1 (Monte Carlo Method)

Usage

```
data(illustration_dist_med_mc)
```

Format

A matrix with 20000 rows and 27 columns:

total Total effect.

direct Direct effect.

indirect Indirect effect.

Author(s)

Ivan Jacob Agaloos Pesigan

```
illustration_dist_med_std
```

Illustration Sampling Distribution Standardized Total, Direct, and Indirect Effects - Time Interval of 1

Description

Illustration Sampling Distribution Standardized Total, Direct, and Indirect Effects - Time Interval of 1

Usage

```
data(illustration_dist_med_std)
```

Format

A matrix with 1000 rows and 27 columns:

total Total effect.

direct Direct effect.

indirect Indirect effect.

Author(s)

Ivan Jacob Agaloos Pesigan

illustration_dist_med_std_mc

Illustration Sampling Distribution Standardized Total, Direct, and Indirect Effects - Time Interval of 1 (Monte Carlo Method)

Description

Illustration Sampling Distribution Standardized Total, Direct, and Indirect Effects - Time Interval of 1 (Monte Carlo Method)

Usage

```
data(illustration_dist_med_std_mc)
```

Format

A matrix with 20000 rows and 27 columns:

total Total effect.

direct Direct effect.

indirect Indirect effect.

Author(s)

Ivan Jacob Agaloos Pesigan

illustration_results *Illustration Small Scale Simulation Results*

Description

Illustration Small Scale Simulation Results

Usage

```
data(illustration_results)
```

Format

A with 22 columns:

taskid Task ID.

replications Number of replications.

effect Total, direct, or indirect effect.

interval Time interval.

parameter Population parameter.

method Method used to generate confidence intervals.

xmy Logical. TRUE for x to m to y path.

std Logical. TRUE for standardized. FALSE for unstandardized.

est Mean parameter estimate.

se Mean standard error.

z Mean z statistic.

p Mean p -value.

R Number of Monte Carlo or bootstrap replications.

ll Mean lower limit of the 95% confidence interval.

ul Mean upper limit of the 95% confidence interval.

sig Proportion of statistically significant results.

zero_hit Proportion of replications where the confidence intervals included zero.

theta_hit Proportion of replications where the confidence intervals included the population parameter.

sq_error Mean squared error.

se_bias Bias in standard error estimate.

coverage Coverage probability.

power Statistical power.

Author(s)

Ivan Jacob Agaloos Pesigan

MCStdXMY

*Monte Carlo Method Confidence Intervals for X-M-Y (Standardized)***Description**

The function generates Monte Carlo method confidence intervals for the mediation model $X \rightarrow M \rightarrow Y$ (Standardized).

Usage

```
MCStdXMY(theta_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- MCStdXMY(theta_hat, seed = 42)
plot(ci)

## End(Not run)
```

MCStdXYM

*Monte Carlo Method Confidence Intervals for X-Y-M (Standardized)***Description**

The function generates Monte Carlo method confidence intervals for the mediation model $X \rightarrow Y \rightarrow M$ (Standardized).

Usage

```
MCStdXYM(theta_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- MCStdXYM(theta_hat, seed = 42)
plot(ci)

## End(Not run)
```

MCStdYMX

*Monte Carlo Method Confidence Intervals for Y-M-X (Standardized)***Description**

The function generates Monte Carlo method confidence intervals for the mediation model $Y \rightarrow M \rightarrow X$ (Standardized).

Usage

```
MCStdYMX(theta_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- MCStdYMX(phi_hat, seed = 42)
plot(ci)

## End(Not run)
```

Description

The function generates Monte Carlo method confidence intervals for the mediation model $X \rightarrow M \rightarrow Y$.

Usage

```
MCXMY(phi_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXYM\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- MCXMY(phi_hat, seed = 42)
plot(ci)

## End(Not run)
```


Description

The function generates Monte Carlo method confidence intervals for the mediation model $X \rightarrow Y \rightarrow M$.

Usage

```
MCXYM(phi_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCYMX\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- MCXYM(phi_hat, seed = 42)
plot(ci)

## End(Not run)
```

MCYMX

*Monte Carlo Method Confidence Intervals for Y-M-X***Description**

The function generates Monte Carlo method confidence intervals for the mediation model $Y \rightarrow M \rightarrow X$.

Usage

```
MCYMX(phi_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: [BootPara\(\)](#), [BootParaStdXMY\(\)](#), [BootParaStdXYM\(\)](#), [BootParaStdYMX\(\)](#), [BootParaXMY\(\)](#), [BootParaXYM\(\)](#), [BootParaYMX\(\)](#), [DeltaStdXMY\(\)](#), [DeltaStdXYM\(\)](#), [DeltaStdYMX\(\)](#), [DeltaXMY\(\)](#), [DeltaXYM\(\)](#), [DeltaYMX\(\)](#), [IllustrationBootPara\(\)](#), [MCStdXMY\(\)](#), [MCStdXYM\(\)](#), [MCStdYMX\(\)](#), [MCXMY\(\)](#), [MCXYM\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- MCYMX(phi_hat, seed = 42)
plot(ci)

## End(Not run)
```

params	<i>Simulation Parameters</i>
--------	------------------------------

Description

Simulation Parameters

Usage

```
data(params)
```

Format

A dataframe with 30 rows and 3 columns:

taskid Simulation Task ID.

n Sample size.

dynamics Dynamics. 0 for original drift matrix, -1 for near-neutral dynamics, and 1 for stronger damping.

Author(s)

Ivan Jacob Agaloos Pesigan

PhiHat	<i>Estimated Drift Matrix</i>
--------	-------------------------------

Description

The function extracts the estimated drift matrix from the fitted model.

Usage

```
PhiHat(fit)
```

Arguments

fit R object. Output of the [FitDynr\(\)](#), [FitMx\(\)](#), [IllustrationFitDynr\(\)](#), or [IllustrationFitMx\(\)](#), functions.

See Also

Other Model Fitting Functions: [FitDynr\(\)](#), [FitMx\(\)](#), [IllustrationFitDynr\(\)](#), [IllustrationFitMx\(\)](#), [IllustrationMCPhiSigma\(\)](#), [IllustrationPrepData\(\)](#), [ThetaHat\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(n = 50)
data <- RandomMeasurement(sim)
fit <- FitDynr(data)
PhiHat(fit)

## End(Not run)
```

RandomMeasurement	<i>Simulate Random Measurement</i>
-------------------	------------------------------------

Description

The function randomly selects 100 observations from the generated data and replaces the unselected observations with NA.

Usage

```
RandomMeasurement(sim)
```

Arguments

`sim` R object. Output of the [GenData\(\)](#) function.

See Also

Other Data Generation Functions: [GenData\(\)](#), [IllustrationGenData\(\)](#)

Examples

```
## Not run:
set.seed(42)
sim <- GenData(taskid = 1)
RandomMeasurement(sim)

## End(Not run)
```

results

*Simulation Results***Description**

Simulation Results

Usage`data(results)`**Format**

A dataframe with 24 columns:

taskid Task ID.**replications** Number of replications.**effect** Total, direct, or indirect effect.**interval** Time interval.**dynamics** Dynamics. 0 for original drift matrix, -1 for near-neutral dynamics, and 1 for stronger damping.**parameter** Population parameter.**method** Method used to generate confidence intervals.**xmy** If TRUE, the mediation model is $X \rightarrow M \rightarrow Y$. If FALSE, the mediation model is $Y \rightarrow M \rightarrow X$.**std** If TRUE, standardized total, direct, and indirect effects. If FALSE, unstandardized total, direct, and indirect effects.**n** Sample size.**est** Mean parameter estimate.**se** Mean standard error.**z** Mean z statistic.**p** Mean p -value.**R** Number of Monte Carlo replications.**ll** Mean lower limit of the 95% confidence interval.**ul** Mean upper limit of the 95% confidence interval.**sig** Proportion of statistically significant results.**zero_hit** Proportion of replications where the confidence intervals contained zero.**theta_hit** Proportion of replications where the confidence intervals contained the population parameter.**sq_error** Mean squared error.**se_bias** Bias in standard error estimate.**coverage** Coverage probability.**power** Statistical power.

Author(s)

Ivan Jacob Agaloos Pesigan

Sim	<i>Simulation Replication</i>
-----	-------------------------------

Description

Simulation Replication

Usage

```
Sim(  
  taskid,  
  repid,  
  output_folder,  
  overwrite,  
  integrity,  
  seed,  
  ci,  
  pb,  
  delta_t,  
  R,  
  B  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
ci	Logical. Run simulations for confidence intervals.
pb	Logical. Run simulations for parametric bootstrap confidence intervals.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.
B	Positive integer. Number of bootstrap samples.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrBootPara	<i>Simulation Replication - BootPara</i>
-----------------	--

Description

Simulation Replication - BootPara

Usage

```
SimDynrBootPara(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity,  
  B,  
  ncores = NULL  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
B	Positive integer. Number of bootstrap samples.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrBootParaStdXMY *Simulation Replication - BootParaStdXMY*

Description

Simulation Replication - BootParaStdXMY

Usage

```
SimDynrBootParaStdXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrBootParaStdYMX *Simulation Replication - BootParaStdYMX*

Description

Simulation Replication - BootParaStdYMX

Usage

```
SimDynrBootParaStdYMX(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrBootParaXMY *Simulation Replication - BootParaXMY*

Description

Simulation Replication - BootParaXMY

Usage

```
SimDynrBootParaXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrBootParaYMX *Simulation Replication - BootParaYMX*

Description

Simulation Replication - BootParaYMX

Usage

```
SimDynrBootParaYMX(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrDeltaStdXMY *Simulation Replication - DynrDeltaStdXMY*

Description

Simulation Replication - DynrDeltaStdXMY

Usage

```
SimDynrDeltaStdXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrDeltaStdYMX	<i>Simulation Replication - DynrDeltaStdYMX</i>
--------------------	---

Description

Simulation Replication - DynrDeltaStdYMX

Usage

```
SimDynrDeltaStdYMX(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity,  
  delta_t  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrDeltaXMY

Simulation Replication - DynrDeltaXMY

Description

Simulation Replication - DynrDeltaXMY

Usage

```
SimDynrDeltaXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrDeltaYMX*Simulation Replication - DynrDeltaYMX*

Description

Simulation Replication - DynrDeltaYMX

Usage

```
SimDynrDeltaYMX(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity,  
  delta_t  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrMCStdXMY	<i>Simulation Replication - DynrMCStdXMY</i>
-----------------	--

Description

Simulation Replication - DynrMCStdXMY

Usage

```
SimDynrMCStdXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimDynrMCStdYMX`*Simulation Replication - DynrMCStdYMX*

Description

Simulation Replication - DynrMCStdYMX

Usage

```
SimDynrMCStdYMX(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity,  
  delta_t,  
  R  
)
```

Arguments

<code>taskid</code>	Positive integer. Task ID.
<code>repid</code>	Positive integer. Replication ID.
<code>output_folder</code>	Character string. Output folder.
<code>seed</code>	Integer. Random seed.
<code>suffix</code>	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
<code>overwrite</code>	Logical. Overwrite existing output in <code>output_folder</code> .
<code>integrity</code>	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
<code>delta_t</code>	Numeric vector. Vector of time intervals.
<code>R</code>	Positive integer. Number of Monte Carlo replications.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrMCXMY

Simulation Replication - DynrMCXMY

Description

Simulation Replication - DynrMCXMY

Usage

```
SimDynrMCXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrMCYMX

Simulation Replication - DynrMCYMX

Description

Simulation Replication - DynrMCYMX

Usage

```
SimDynrMCYMX(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitDynr	<i>Simulation Replication - FitDynr</i>
------------	---

Description

Simulation Replication - FitDynr

Usage

```
SimFitDynr(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMx	<i>Simulation Replication - FitMx</i>
----------	---------------------------------------

Description

Simulation Replication - FitMx

Usage

SimFitMx(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN	<i>Simulation File Name</i>
-------	-----------------------------

Description

Simulation File Name

Usage

SimFN(output_type, output_folder, suffix)

Arguments

- output_type Character string. Output type.
- output_folder Character string. Output folder.
- suffix Character string. Output of manCTMed:::SimSuffix().

Value

Returns a character string file name with the output_folder in the OS-specific format.

SimGenData	<i>Simulation Replication - GenData</i>
------------	---

Description

Simulation Replication - GenData

Usage

SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

- taskid Positive integer. Task ID.
- repid Positive integer. Replication ID.
- output_folder Character string. Output folder.
- seed Integer. Random seed.
- suffix Character string. Output of manCTMed:::SimSuffix().
- overwrite Logical. Overwrite existing output in output_folder.
- integrity Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustration	<i>Simulation Replication (Illustration)</i>
-----------------	--

Description

Simulation Replication (Illustration)

Usage

```
SimIllustration(  
  taskid,  
  repid,  
  output_folder,  
  overwrite,  
  integrity,  
  seed,  
  ci,  
  pb,  
  delta_t,  
  R,  
  B  
)
```

Arguments

- | | |
|---------------|---|
| taskid | Positive integer. Task ID. |
| repid | Positive integer. Replication ID. |
| output_folder | Character string. Output folder. |
| overwrite | Logical. Overwrite existing output in output_folder. |
| integrity | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| seed | Integer. Random seed. |
| ci | Logical. Run simulations for confidence intervals. |
| pb | Logical. Run simulations for parametric bootstrap confidence intervals. |

delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.
B	Positive integer. Number of bootstrap samples.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustrationDynrBootPara

Simulation Replication - BootPara

Description

Simulation Replication - BootPara

Usage

```
SimIllustrationDynrBootPara(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  B,
  ncores = NULL
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
B	Positive integer. Number of bootstrap samples.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the IllustrationSim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustrationDynrBootParaStdXMY
<i>Simulation Replication - BootParaStdXMY</i>

Description

Simulation Replication - BootParaStdXMY

Usage

```
SimIllustrationDynrBootParaStdXMY(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity,  
  delta_t  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the IllustrationSim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustrationDynrBootParaXMY
<i>Simulation Replication - BootParaXMY</i>

Description

Simulation Replication - BootParaXMY

Usage

```
SimIllustrationDynrBootParaXMY(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity,  
  delta_t  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the IllustrationSim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustrationDynrDeltaStdXMY
<i>Simulation Replication - Illustration (DynrDeltaStdXMY)</i>

Description

Simulation Replication - Illustration (DynrDeltaStdXMY)

Usage

```
SimIllustrationDynrDeltaStdXMY(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity,  
  delta_t  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the IllustrationSim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustrationDynrDeltaXMY
<i>Simulation Replication - Illustration (DynrDeltaXMY)</i>

Description

Simulation Replication - Illustration (DynrDeltaXMY)

Usage

```
SimIllustrationDynrDeltaXMY(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity,  
  delta_t  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the IllustrationSim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustrationDynrMCPhiSigma
<i>Simulation Replication - Illustration (MCPhiSigma)</i>

Description

Simulation Replication - Illustration (MCPhiSigma)

Usage

```
SimIllustrationDynrMCPhiSigma(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity,  
  R  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
R	Positive integer. Number of Monte Carlo replications.

Details

This function is executed via the IllustrationSim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustrationDynrMCStdXMY

Simulation Replication - Illustration (DynrMCStdXMY)

Description

Simulation Replication - Illustration (DynrMCStdXMY)

Usage

```
SimIllustrationDynrMCStdXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.

Details

This function is executed via the IllustrationSim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustrationDynrMCXMY

Simulation Replication - Illustration (DynrMCXMY)

Description

Simulation Replication - Illustration (DynrMCXMY)

Usage

```
SimIllustrationDynrMCXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of Monte Carlo replications.

Details

This function is executed via the `IllustrationSim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimIllustrationFitDynr`

Simulation Replication - IllustrationFitDynr

Description

Simulation Replication - IllustrationFitDynr

Usage

```
SimIllustrationFitDynr(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity
)
```

Arguments

<code>taskid</code>	Positive integer. Task ID.
<code>repid</code>	Positive integer. Replication ID.
<code>output_folder</code>	Character string. Output folder.
<code>seed</code>	Integer. Random seed.
<code>suffix</code>	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
<code>overwrite</code>	Logical. Overwrite existing output in <code>output_folder</code> .
<code>integrity</code>	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `IllustrationSim` function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustrationFitMx *Simulation Replication - IllustrationFitMx*

Description

Simulation Replication - IllustrationFitMx

Usage

```
SimIllustrationFitMx(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the IllustrationSim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustrationGenData

Simulation Replication - IllustrationGenData

Description

Simulation Replication - IllustrationGenData

Usage

```
SimIllustrationGenData(  
  taskid,  
  repid,  
  output_folder,  
  seed,  
  suffix,  
  overwrite,  
  integrity  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `IllustrationSim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimIllustrationPara	<i>Simulation Replication Parametric Bootstrap (Parallel)</i>
---------------------	---

Description

Simulation Replication Parametric Bootstrap (Parallel)

Usage

```
SimIllustrationPara(  
  taskid,  
  repid,  
  output_folder,  
  overwrite,  
  integrity,  
  seed,  
  B  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
B	Positive integer. Number of bootstrap samples.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimPara

Simulation Replication Parametric Bootstrap (Parallel)

Description

Simulation Replication Parametric Bootstrap (Parallel)

Usage

```
SimPara(taskid, repid, output_folder, overwrite, integrity, seed, B)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
B	Positive integer. Number of bootstrap samples.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimProj

Simulation Project Name

Description

Simulation Project Name

Usage

```
SimProj()
```

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

Sum	<i>Summary</i>
-----	----------------

Description

Summary

Usage

Sum(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumDynrDeltaStdXMY	<i>Summary (DynrDeltaStdXMY)</i>
--------------------	----------------------------------

Description

Summary (DynrDeltaStdXMY)

Usage

SumDynrDeltaStdXMY(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumDynrDeltaStdYMX	<i>Summary (DynrDeltaStdYMX)</i>
--------------------	----------------------------------

Description

Summary (DynrDeltaStdYMX)

Usage

SumDynrDeltaStdYMX(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumDynrDeltaXMY	<i>Summary (DynrDeltaXMY)</i>
-----------------	-------------------------------

Description

Summary (DynrDeltaXMY)

Usage

SumDynrDeltaXMY(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumDynrDeltaYMX	<i>Summary (DynrDeltaYMX)</i>
-----------------	-------------------------------

Description

Summary (DynrDeltaYMX)

Usage

SumDynrDeltaYMX(taskid, reps, output_folder, overwrite, integrity)

Arguments

- taskid Positive integer. Task ID.
- reps Positive integer. Number of replications.
- output_folder Character string. Output folder.
- overwrite Logical. Overwrite existing output in output_folder.
- integrity Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumDynrMCStdXMY	<i>Summary (DynrMCStdXMY)</i>
-----------------	-------------------------------

Description

Summary (DynrMCStdXMY)

Usage

SumDynrMCStdXMY(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumDynrMCStdYMX	<i>Summary (DynrMCStdYMX)</i>
-----------------	-------------------------------

Description

Summary (DynrMCStdYMX)

Usage

SumDynrMCStdYMX(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumDynrMCXMY	<i>Summary (DynrMCXMY)</i>
--------------	----------------------------

Description

Summary (DynrMCXMY)

Usage

SumDynrMCXMY(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumDynrMCYMX	<i>Summary (DynrMCYMX)</i>
--------------	----------------------------

Description

Summary (DynrMCYMX)

Usage

SumDynrMCYMX(taskid, reps, output_folder, overwrite, integrity)

Arguments

- taskid Positive integer. Task ID.
- reps Positive integer. Number of replications.
- output_folder Character string. Output folder.
- overwrite Logical. Overwrite existing output in output_folder.
- integrity Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitDynr	<i>Summary (FitDynr)</i>
------------	--------------------------

Description

Summary (FitDynr)

Usage

SumFitDynr(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumIllustration	<i>Summary (Illustration)</i>
-----------------	-------------------------------

Description

Summary (Illustration)

Usage

SumIllustration(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumIllustrationDynrBootParaStdXMY
<i>Summary - Illustration (DynrBootParaStdXMY)</i>

Description

Summary - Illustration (DynrBootParaStdXMY)

Usage

```
SumIllustrationDynrBootParaStdXMY(  
    taskid,  
    reps,  
    output_folder,  
    overwrite,  
    integrity,  
    type = "pc"  
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
type	Character string. Confidence interval type.

Details

This function is executed via the IllustrationSum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumIllustrationDynrBootParaXMY

Summary - Illustration (DynrBootParaXMY)

Description

Summary - Illustration (DynrBootParaXMY)

Usage

```
SumIllustrationDynrBootParaXMY(  
  taskid,  
  reps,  
  output_folder,  
  overwrite,  
  integrity,  
  type = "pc"  
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
type	Character string. Confidence interval type.

Details

This function is executed via the IllustrationSum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumIllustrationDynrDeltaStdXMY
<i>Summary - Illustration (DynrDeltaStdXMY)</i>

Description

Summary - Illustration (DynrDeltaStdXMY)

Usage

```
SumIllustrationDynrDeltaStdXMY(  
    taskid,  
    reps,  
    output_folder,  
    overwrite,  
    integrity  
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the IllustrationSum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumIllustrationDynrDeltaXMY
<i>Summary - Illustration (DynrDeltaXMY)</i>

Description

Summary - Illustration (DynrDeltaXMY)

Usage

SumIllustrationDynrDeltaXMY(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the IllustrationSum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumIllustrationDynrMCStdXMY
<i>Summary - Illustration (DynrMCStdXMY)</i>

Description

Summary - Illustration (DynrMCStdXMY)

Usage

SumIllustrationDynrMCStdXMY(taskid, reps, output_folder, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the IllustrationSum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumIllustrationDynrMCXMY

Summary - Illustration (DynrMCXMY)

Description

Summary - Illustration (DynrMCXMY)

Usage

```
SumIllustrationDynrMCXMY(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the IllustrationSum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumIllustrationFitDynr

Summary - Illustration (FitDynr)

Description

Summary - Illustration (FitDynr)

Usage

```
SumIllustrationFitDynr(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the IllustrationSum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

ThetaHat

*Estimated Drift Matrix and Process Noise***Description**

The function extracts the estimated drift matrix and process noise from the fitted model.

Usage

```
ThetaHat(fit)
```

Arguments

`fit` R object. Output of the [FitDynr\(\)](#), [FitMx\(\)](#), [IllustrationFitDynr\(\)](#), or [IllustrationFitMx\(\)](#), functions.

See Also

Other Model Fitting Functions: [FitDynr\(\)](#), [FitMx\(\)](#), [IllustrationFitDynr\(\)](#), [IllustrationFitMx\(\)](#), [IllustrationMCPhiSigma\(\)](#), [IllustrationPrepData\(\)](#), [PhiHat\(\)](#)

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(n = 50)
data <- RandomMeasurement(sim)
fit <- FitDynr(data)
ThetaHat(fit)

## End(Not run)
```

Index

* Compression Functions

Compress, [11](#)
 IllustrationCompress, [24](#)

* Confidence Interval Functions

BootPara, [4](#)
 BootParaStdXMY, [5](#)
 BootParaStdXYM, [6](#)
 BootParaStdYMX, [7](#)
 BootParaXMY, [8](#)
 BootParaXYM, [9](#)
 BootParaYMX, [10](#)
 DeltaStdXMY, [12](#)
 DeltaStdXYM, [12](#)
 DeltaStdYMX, [13](#)
 DeltaXMY, [14](#)
 DeltaXYM, [15](#)
 DeltaYMX, [16](#)
 IllustrationBootPara, [23](#)
 MCStdXMY, [37](#)
 MCStdXYM, [38](#)
 MCStdYMX, [39](#)
 MCXMY, [40](#)
 MCXYM, [41](#)
 MCYMX, [42](#)

* Data Generation Functions

GenData, [22](#)
 IllustrationGenData, [29](#)
 RandomMeasurement, [44](#)

* Figure Functions

FigPlotEffects, [16](#)
 FigScatterPlotCoverage, [17](#)
 FigScatterPlotPower, [18](#)
 FigScatterPlotSeBias, [19](#)
 FigScatterPlotType1, [20](#)
 IllustrationFigPlotEffects, [25](#)
 IllustrationFigScatterPlotCoverage, [25](#)
 IllustrationFigScatterPlotPower, [26](#)

IllustrationFigScatterPlotSeBias, [27](#)

* Model Fitting Functions

FitDynr, [21](#)
 FitMx, [21](#)
 IllustrationFitDynr, [28](#)
 IllustrationFitMx, [28](#)
 IllustrationMCPhiSigma, [30](#)
 IllustrationPrepData, [30](#)
 PhiHat, [43](#)
 ThetaHat, [91](#)

* ci

BootPara, [4](#)
 BootParaStdXMY, [5](#)
 BootParaStdXYM, [6](#)
 BootParaStdYMX, [7](#)
 BootParaXMY, [8](#)
 BootParaXYM, [9](#)
 BootParaYMX, [10](#)
 DeltaStdXMY, [12](#)
 DeltaStdXYM, [12](#)
 DeltaStdYMX, [13](#)
 DeltaXMY, [14](#)
 DeltaXYM, [15](#)
 DeltaYMX, [16](#)
 IllustrationBootPara, [23](#)
 IllustrationMCPhiSigma, [30](#)
 MCStdXMY, [37](#)
 MCStdXYM, [38](#)
 MCStdYMX, [39](#)
 MCXMY, [40](#)
 MCXYM, [41](#)
 MCYMX, [42](#)
 PhiHat, [43](#)
 SimDynrBootPara, [47](#)
 SimDynrBootParaStdXMY, [48](#)
 SimDynrBootParaStdYMX, [49](#)
 SimDynrBootParaXMY, [50](#)
 SimDynrBootParaYMX, [51](#)

- SimDynrDeltaStdXMY, [52](#)
- SimDynrDeltaStdYMX, [53](#)
- SimDynrDeltaXMY, [54](#)
- SimDynrDeltaYMX, [55](#)
- SimDynrMCStdXMY, [56](#)
- SimDynrMCStdYMX, [57](#)
- SimDynrMCXMY, [58](#)
- SimDynrMCYMX, [59](#)
- SimIllustrationDynrBootPara, [64](#)
- SimIllustrationDynrBootParaStdXMY, [65](#)
- SimIllustrationDynrBootParaXMY, [66](#)
- SimIllustrationDynrDeltaStdXMY, [67](#)
- SimIllustrationDynrDeltaXMY, [68](#)
- SimIllustrationDynrMCPhiSigma, [69](#)
- SimIllustrationDynrMCStdXMY, [70](#)
- SimIllustrationDynrMCXMY, [71](#)
- ThetaHat, [91](#)
- * **compress**
 - Compress, [11](#)
 - IllustrationCompress, [24](#)
- * **data**
 - illustration_dist, [31](#)
 - illustration_dist_dt, [32](#)
 - illustration_dist_dt_mc, [32](#)
 - illustration_dist_mc, [33](#)
 - illustration_dist_med, [33](#)
 - illustration_dist_med_mc, [34](#)
 - illustration_dist_med_std, [34](#)
 - illustration_dist_med_std_mc, [35](#)
 - illustration_results, [35](#)
 - params, [43](#)
 - results, [45](#)
- * **figure**
 - FigPlotEffects, [16](#)
 - FigScatterPlotCoverage, [17](#)
 - FigScatterPlotPower, [18](#)
 - FigScatterPlotSeBias, [19](#)
 - FigScatterPlotType1, [20](#)
 - IllustrationFigPlotEffects, [25](#)
 - IllustrationFigScatterPlotCoverage, [25](#)
 - IllustrationFigScatterPlotPower, [26](#)
 - IllustrationFigScatterPlotSeBias, [27](#)
- * **fit**
 - FitDynr, [21](#)
 - FitMx, [21](#)
 - IllustrationFitDynr, [28](#)
 - IllustrationFitMx, [28](#)
 - SimFitDynr, [60](#)
 - SimFitMx, [61](#)
 - SimIllustrationFitDynr, [72](#)
 - SimIllustrationFitMx, [73](#)
- * **gendata**
 - GenData, [22](#)
 - IllustrationGenData, [29](#)
 - IllustrationPrepData, [30](#)
 - RandomMeasurement, [44](#)
 - SimGenData, [62](#)
 - SimIllustrationGenData, [74](#)
- * **illustration**
 - illustration_dist, [31](#)
 - illustration_dist_dt, [32](#)
 - illustration_dist_dt_mc, [32](#)
 - illustration_dist_mc, [33](#)
 - illustration_dist_med, [33](#)
 - illustration_dist_med_mc, [34](#)
 - illustration_dist_med_std, [34](#)
 - illustration_dist_med_std_mc, [35](#)
 - illustration_results, [35](#)
 - IllustrationBootPara, [23](#)
 - IllustrationFigScatterPlotCoverage, [25](#)
 - IllustrationFigScatterPlotPower, [26](#)
 - IllustrationFigScatterPlotSeBias, [27](#)
 - IllustrationFitDynr, [28](#)
 - IllustrationFitMx, [28](#)
 - IllustrationGenData, [29](#)
 - IllustrationMCPhiSigma, [30](#)
 - IllustrationPrepData, [30](#)
 - SimIllustration, [63](#)
 - SimIllustrationDynrBootPara, [64](#)
 - SimIllustrationDynrBootParaStdXMY, [65](#)
 - SimIllustrationDynrBootParaXMY, [66](#)
 - SimIllustrationDynrDeltaStdXMY, [67](#)
 - SimIllustrationDynrDeltaXMY, [68](#)
 - SimIllustrationDynrMCPhiSigma, [69](#)
 - SimIllustrationDynrMCStdXMY, [70](#)
 - SimIllustrationDynrMCXMY, [71](#)
 - SimIllustrationFitDynr, [72](#)
 - SimIllustrationFitMx, [73](#)

- SimIllustrationGenData, [74](#)
- SimIllustrationPara, [75](#)
- SumIllustration, [84](#)
- SumIllustrationDynrBootParaStdXMY, [85](#)
- SumIllustrationDynrBootParaXMY, [86](#)
- SumIllustrationDynrDeltaStdXMY, [87](#)
- SumIllustrationDynrDeltaXMY, [88](#)
- SumIllustrationDynrMCStdXMY, [88](#)
- SumIllustrationDynrMCXMY, [89](#)
- SumIllustrationFitDynr, [90](#)
- * **manCTMed**
 - BootPara, [4](#)
 - BootParaStdXMY, [5](#)
 - BootParaStdXYM, [6](#)
 - BootParaStdYMX, [7](#)
 - BootParaXMY, [8](#)
 - BootParaXYM, [9](#)
 - BootParaYMX, [10](#)
 - Compress, [11](#)
 - DeltaStdXMY, [12](#)
 - DeltaStdXYM, [12](#)
 - DeltaStdYMX, [13](#)
 - DeltaXMY, [14](#)
 - DeltaXYM, [15](#)
 - DeltaYMX, [16](#)
 - FigPlotEffects, [16](#)
 - FigScatterPlotCoverage, [17](#)
 - FigScatterPlotPower, [18](#)
 - FigScatterPlotSeBias, [19](#)
 - FigScatterPlotType1, [20](#)
 - FitDynr, [21](#)
 - FitMx, [21](#)
 - GenData, [22](#)
 - IllustrationBootPara, [23](#)
 - IllustrationCompress, [24](#)
 - IllustrationFigPlotEffects, [25](#)
 - IllustrationFigScatterPlotCoverage, [25](#)
 - IllustrationFigScatterPlotPower, [26](#)
 - IllustrationFigScatterPlotSeBias, [27](#)
 - IllustrationFitDynr, [28](#)
 - IllustrationFitMx, [28](#)
 - IllustrationGenData, [29](#)
 - IllustrationMCPhiSigma, [30](#)
 - IllustrationPrepData, [30](#)
 - MCStdXMY, [37](#)
 - MCStdXYM, [38](#)
 - MCStdYMX, [39](#)
 - MCXMY, [40](#)
 - MCXYM, [41](#)
 - MCYMX, [42](#)
 - PhiHat, [43](#)
 - RandomMeasurement, [44](#)
 - Sim, [46](#)
 - SimDynrBootPara, [47](#)
 - SimDynrBootParaStdXMY, [48](#)
 - SimDynrBootParaStdYMX, [49](#)
 - SimDynrBootParaXMY, [50](#)
 - SimDynrBootParaYMX, [51](#)
 - SimDynrDeltaStdXMY, [52](#)
 - SimDynrDeltaStdYMX, [53](#)
 - SimDynrDeltaXMY, [54](#)
 - SimDynrDeltaYMX, [55](#)
 - SimDynrMCStdXMY, [56](#)
 - SimDynrMCStdYMX, [57](#)
 - SimDynrMCXMY, [58](#)
 - SimDynrMCYMX, [59](#)
 - SimFitDynr, [60](#)
 - SimFitMx, [61](#)
 - SimFN, [62](#)
 - SimGenData, [62](#)
 - SimIllustration, [63](#)
 - SimIllustrationDynrBootPara, [64](#)
 - SimIllustrationDynrBootParaStdXMY, [65](#)
 - SimIllustrationDynrBootParaXMY, [66](#)
 - SimIllustrationDynrDeltaStdXMY, [67](#)
 - SimIllustrationDynrDeltaXMY, [68](#)
 - SimIllustrationDynrMCPhiSigma, [69](#)
 - SimIllustrationDynrMCStdXMY, [70](#)
 - SimIllustrationDynrMCXMY, [71](#)
 - SimIllustrationFitDynr, [72](#)
 - SimIllustrationFitMx, [73](#)
 - SimIllustrationGenData, [74](#)
 - SimIllustrationPara, [75](#)
 - SimPara, [76](#)
 - SimProj, [76](#)
 - Sum, [77](#)
 - SumDynrDeltaStdXMY, [77](#)
 - SumDynrDeltaStdYMX, [78](#)
 - SumDynrDeltaXMY, [79](#)
 - SumDynrDeltaYMX, [80](#)
 - SumDynrMCStdXMY, [80](#)

- SumDynrMCStdYMX, 81
- SumDynrMCXMY, 82
- SumDynrMCYMX, 83
- SumFitDynr, 83
- SumIllustration, 84
- SumIllustrationDynrBootParaStdXMY, 85
- SumIllustrationDynrBootParaXMY, 86
- SumIllustrationDynrDeltaStdXMY, 87
- SumIllustrationDynrDeltaXMY, 88
- SumIllustrationDynrMCStdXMY, 88
- SumIllustrationDynrMCXMY, 89
- SumIllustrationFitDynr, 90
- ThetaHat, 91
- * **parameters**
 - params, 43
 - results, 45
- * **simulation**
 - Sim, 46
 - SimDynrBootPara, 47
 - SimDynrBootParaStdXMY, 48
 - SimDynrBootParaStdYMX, 49
 - SimDynrBootParaXMY, 50
 - SimDynrBootParaYMX, 51
 - SimDynrDeltaStdXMY, 52
 - SimDynrDeltaStdYMX, 53
 - SimDynrDeltaXMY, 54
 - SimDynrDeltaYMX, 55
 - SimDynrMCStdXMY, 56
 - SimDynrMCStdYMX, 57
 - SimDynrMCXMY, 58
 - SimDynrMCYMX, 59
 - SimFitDynr, 60
 - SimFitMx, 61
 - SimFN, 62
 - SimGenData, 62
 - SimPara, 76
 - SimProj, 76
 - Sum, 77
 - SumDynrDeltaStdXMY, 77
 - SumDynrDeltaStdYMX, 78
 - SumDynrDeltaXMY, 79
 - SumDynrDeltaYMX, 80
 - SumDynrMCStdXMY, 80
 - SumDynrMCStdYMX, 81
 - SumDynrMCXMY, 82
 - SumDynrMCYMX, 83
 - SumFitDynr, 83
 - SumIllustration, 84
 - SumIllustrationDynrBootParaStdXMY, 85
 - SumIllustrationDynrBootParaXMY, 86
 - SumIllustrationDynrDeltaStdXMY, 87
 - SumIllustrationDynrDeltaXMY, 88
 - SumIllustrationDynrMCStdXMY, 88
 - SumIllustrationDynrMCXMY, 89
 - SumIllustrationFitDynr, 90
- * **summary**
 - Sum, 77
 - SumDynrDeltaStdXMY, 77
 - SumDynrDeltaStdYMX, 78
 - SumDynrDeltaXMY, 79
 - SumDynrDeltaYMX, 80
 - SumDynrMCStdXMY, 80
 - SumDynrMCStdYMX, 81
 - SumDynrMCXMY, 82
 - SumDynrMCYMX, 83
 - SumFitDynr, 83
 - SumIllustration, 84
 - SumIllustrationDynrBootParaStdXMY, 85
 - SumIllustrationDynrBootParaXMY, 86
 - SumIllustrationDynrDeltaStdXMY, 87
 - SumIllustrationDynrDeltaXMY, 88
 - SumIllustrationDynrMCStdXMY, 88
 - SumIllustrationDynrMCXMY, 89
 - SumIllustrationFitDynr, 90
- BootPara, 4
- BootPara(), 5–10, 12–16, 23, 37–42
- BootParaStdXMY, 5
- BootParaStdXMY(), 5–10, 12–16, 23, 37–42
- BootParaStdXYM, 6
- BootParaStdXYM(), 5, 7–10, 12–16, 23, 37–42
- BootParaStdYMX, 7
- BootParaStdYMX(), 5, 6, 8–10, 12–16, 23, 37–42
- BootParaXMY, 8
- BootParaXMY(), 5–7, 9, 10, 12–16, 23, 37–42
- BootParaXYM, 9
- BootParaXYM(), 5–8, 10, 12–16, 23, 37–42
- BootParaYMX, 10
- BootParaYMX(), 5–9, 12–16, 23, 37–42
- Compress, 11
- Compress(), 24
- DeltaStdXMY, 12
- DeltaStdXMY(), 5–10, 13–16, 23, 37–42
- DeltaStdXYM, 12
- DeltaStdXYM(), 5–10, 12, 14–16, 23, 37–42
- DeltaStdYMX, 13
- DeltaStdYMX(), 5–10, 12–16, 23, 37–42
- DeltaXMY, 14
- DeltaXMY(), 5–10, 12–16, 23, 37–42

- DeltaXYM, 15
- DeltaXYM(), 5–10, 12–14, 16, 23, 37–42
- DeltaYMX, 16
- DeltaYMX(), 5–10, 12–15, 23, 37–42
- dynr, 21, 28

- FigPlotEffects, 16
- FigPlotEffects(), 18–20, 25–27
- FigScatterPlotCoverage, 17
- FigScatterPlotCoverage(), 17–20, 25–27
- FigScatterPlotPower, 18
- FigScatterPlotPower(), 17–20, 25–27
- FigScatterPlotSeBias, 19
- FigScatterPlotSeBias(), 17, 18, 20, 25–27
- FigScatterPlotType1, 20
- FigScatterPlotType1(), 17–19, 25–27
- FitDynr, 21
- FitDynr(), 4, 22, 28–31, 43, 91
- FitMx, 21
- FitMx(), 4, 21, 28–31, 43, 91

- GenData, 22
- GenData(), 29, 44

- illustration_dist, 31
- illustration_dist_dt, 32
- illustration_dist_dt_mc, 32
- illustration_dist_mc, 33
- illustration_dist_med, 33
- illustration_dist_med_mc, 34
- illustration_dist_med_std, 34
- illustration_dist_med_std_mc, 35
- illustration_results, 35
- IllustrationBootPara, 23
- IllustrationBootPara(), 5–10, 12–16, 37–42
- IllustrationCompress, 24
- IllustrationCompress(), 11
- IllustrationFigPlotEffects, 25
- IllustrationFigPlotEffects(), 17–20, 26, 27
- IllustrationFigScatterPlotCoverage, 25
- IllustrationFigScatterPlotCoverage(), 17–20, 25–27
- IllustrationFigScatterPlotPower, 26
- IllustrationFigScatterPlotPower(), 17–20, 25–27
- IllustrationFigScatterPlotSeBias, 27
- IllustrationFigScatterPlotSeBias(), 17–20, 25, 26
- IllustrationFitDynr, 28
- IllustrationFitDynr(), 4, 21, 22, 29–31, 43, 91
- IllustrationFitMx, 28
- IllustrationFitMx(), 4, 21, 22, 28, 30, 31, 43, 91
- IllustrationGenData, 29
- IllustrationGenData(), 22, 30, 31, 44
- IllustrationMCPhiSigma, 30
- IllustrationMCPhiSigma(), 21, 22, 28, 29, 31, 43, 91
- IllustrationPrepData, 30
- IllustrationPrepData(), 21, 22, 28–30, 43, 91

- MCStdXMY, 37
- MCStdXMY(), 5–10, 12–16, 23, 38–42
- MCStdXYM, 38
- MCStdXYM(), 5–10, 12–16, 23, 37, 39–42
- MCStdYMX, 39
- MCStdYMX(), 5–10, 12–16, 23, 37, 38, 40–42
- MCXMY, 40
- MCXMY(), 5–10, 12–16, 23, 37–39, 41, 42
- MCXYM, 41
- MCXYM(), 5–10, 12–16, 23, 37–40, 42
- MCYMX, 42
- MCYMX(), 5–10, 12–16, 23, 37–41

- OpenMx, 21, 28

- params, 43
- PhiHat, 43
- PhiHat(), 8–10, 14–16, 21, 22, 28–31, 40–42, 91

- RandomMeasurement, 44
- RandomMeasurement(), 21, 22, 29
- results, 45

- Sim, 46
- SimDynrBootPara, 47
- SimDynrBootParaStdXMY, 48
- SimDynrBootParaStdYMX, 49
- SimDynrBootParaXMY, 50
- SimDynrBootParaYMX, 51
- SimDynrDeltaStdXMY, 52
- SimDynrDeltaStdYMX, 53

SimDynrDeltaXMY, 54
 SimDynrDeltaYMX, 55
 SimDynrMCStdXMY, 56
 SimDynrMCStdYMX, 57
 SimDynrMCXMY, 58
 SimDynrMCYMX, 59
 SimFitDynr, 60
 SimFitMx, 61
 SimFN, 62
 SimGenData, 62
 SimIllustration, 63
 SimIllustrationDynrBootPara, 64
 SimIllustrationDynrBootParaStdXMY, 65
 SimIllustrationDynrBootParaXMY, 66
 SimIllustrationDynrDeltaStdXMY, 67
 SimIllustrationDynrDeltaXMY, 68
 SimIllustrationDynrMCPhiSigma, 69
 SimIllustrationDynrMCStdXMY, 70
 SimIllustrationDynrMCXMY, 71
 SimIllustrationFitDynr, 72
 SimIllustrationFitMx, 73
 SimIllustrationGenData, 74
 SimIllustrationPara, 75
 SimPara, 76
 SimProj, 76
 simStateSpace::SimSSMOUFixed(), 22, 29
 Sum, 77
 SumDynrDeltaStdXMY, 77
 SumDynrDeltaStdYMX, 78
 SumDynrDeltaXMY, 79
 SumDynrDeltaYMX, 80
 SumDynrMCStdXMY, 80
 SumDynrMCStdYMX, 81
 SumDynrMCXMY, 82
 SumDynrMCYMX, 83
 SumFitDynr, 83
 SumIllustration, 84
 SumIllustrationDynrBootParaStdXMY, 85
 SumIllustrationDynrBootParaXMY, 86
 SumIllustrationDynrDeltaStdXMY, 87
 SumIllustrationDynrDeltaXMY, 88
 SumIllustrationDynrMCStdXMY, 88
 SumIllustrationDynrMCXMY, 89
 SumIllustrationFitDynr, 90

 ThetaHat, 91
 ThetaHat(), 5–7, 12, 13, 21, 22, 28–31,
 37–39, 43